

Objective

To implement and demonstrate `SecurityException` in Java using try-catch by explicitly throwing a security-related exception.

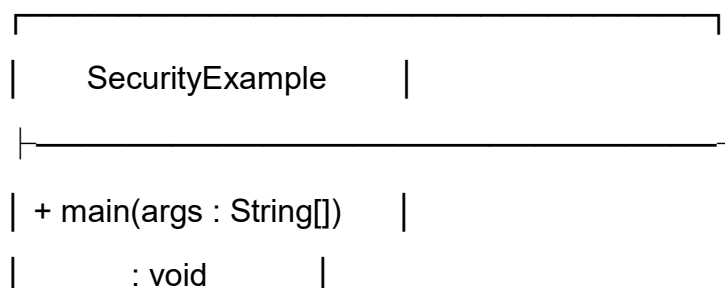
Steps for Execution

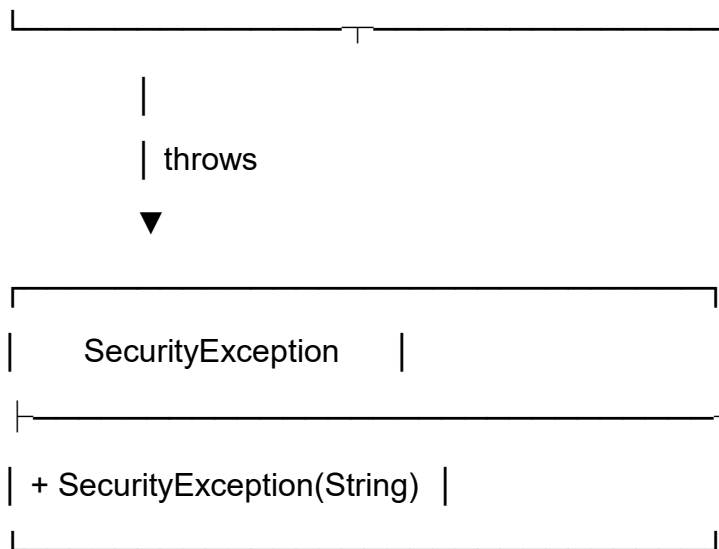
1. Create a Java project and package named `javacore`.
2. Create a file named `SecurityExample.java` and enter the given code.
3. Use a try block to check security.
4. Explicitly throw a `SecurityException`.
5. Handle the exception using catch.
6. Compile and run the program.
7. Observe the output.

Algorithm

1. Start the program.
2. Enter the try block.
3. Display the security checking message.
4. Throw a `SecurityException`.
5. Catch the exception.
6. Display the error message.
7. Display "Program ended.".
8. Stop the program.

UML Class Diagram





Relationship

- `SecurityExample` explicitly throws a `SecurityException`.
- The exception represents a security or access restriction.
- The catch block handles the `SecurityException`.

CODE:

```
package javacore;

public class SecurityExample {

    public static void main(String[] args) {

        try {

            System.out.println("Checking security...");

            throw new SecurityException("Access denied");

        }

        catch (SecurityException e) {

            System.out.println("Error: Security permission denied.");

        }

    }

}
```

```
        System.out.println("Program ended.");  
    }  
}
```

Output

Checking security...

Error: Security permission denied.

Program ended.

Result

The program successfully demonstrates `SecurityException` handling in Java. A `SecurityException` is explicitly thrown when access is denied and is handled using the catch block, allowing the program to continue normally.